

# SYDNEY TECHNICAL HIGH SCHOOL

(Est 1911)



## MATHEMATICS

### YEAR 11 YEARLY EXAMINATION

SEPTEMBER 2004

Time allowed : 2 hours

#### Instructions :

- Show all necessary working in every question.
- Start each question on a new page.
- Attempt all questions.
- All questions are of equal value.
- Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- These questions are to be handed in with your answers.
- Answers must be written in blue or black pen.

Name : \_\_\_\_\_

Class : \_\_\_\_\_

| Question<br>1 | Question<br>2 | Question<br>3 | Question<br>4 | Question<br>5 | Question<br>6 | Question<br>7 | Question<br>8 | TOTAL |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|-------|
|               |               |               |               |               |               |               |               |       |

### Question 1

- a) Find the midpoint of the interval from  $A(-3,6)$  to  $B(4,0)$ . 1
- b) Factorise fully  $12a^3 - 4a^2 - a$  2
- c) Find  $\frac{d}{dx}(6x^2 - 3x - 1)$  1
- d) Find the exact value of  $\sin 240^\circ$  1
- e) Solve  $|x - 3| = |2x + 1|$  2
- f) Lenny paid \$207 for a meal in a restaurant. This included a 15% tip.  
What was the cost of the meal without the tip? 2
- g) Solve  $6x^2 = x$  2

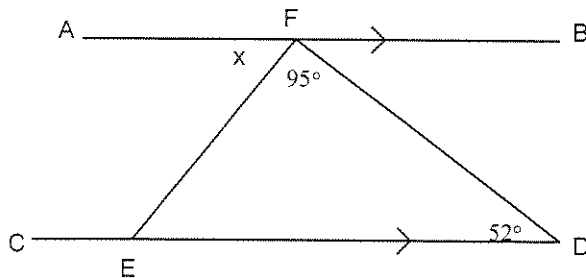
**Question 2** (Start a new page)

- a) Evaluate the following correct to 3 significant figures. 2

$$\sqrt{\frac{4^2 + 18^2}{23^2 - 18^2}}$$

- b) Solve  $x - \frac{x+4}{4} = 2$  2

- c) Find  $x$ , giving reasons 2



- d) i) Sketch  $y = \sqrt{9 - x^2}$  2

- ii) State the range of  $y = \sqrt{9 - x^2}$  1

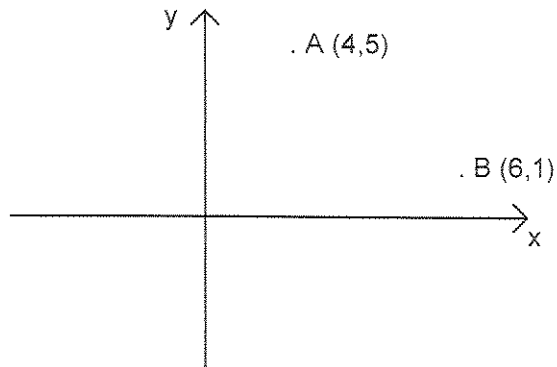
- e) Solve simultaneously for  $x$  and  $y$  2

$$2x - y = 7$$

$$x - 2y = -1$$

**Question 3** (Start a new page)

a)



- i) Show that the equation of the line AB is given by 2

$$2x + y - 13 = 0$$

- ii) The line AB is produced to meet the  $x$  axis at T. 1

Find the coordinates of T.

- iii) Find the distance AB 1

- iv) Find the perpendicular distance of the point  $(-1, 5)$  2

to the line AB.

- v) ABC is a right angled isosceles triangle with  $\angle BAC = 90^\circ$  2

Find all the possible coordinates of the point C.

- b) Sketch the region satisfied simultaneously by the inequalities 3

$$x + y + 2 \leq 0 \quad \text{and} \quad x - 2y \geq 0$$

( Your diagram must be neat and clearly labelled )

**Question 4** (Start a new page)

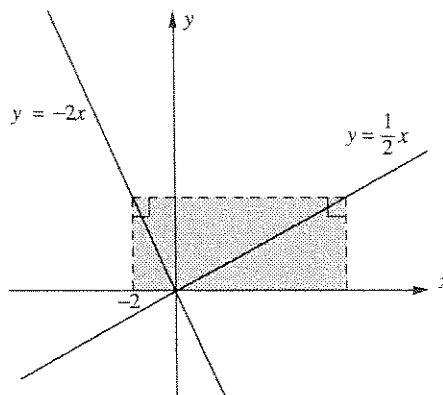
- a) Find the equation of the line that passes through the point  $(2, 4)$  and is parallel to the  $x$  axis. 1

- b) If  $\sin \theta = \frac{1}{5}$  and  $\theta$  is obtuse, find the exact value of  $\cos \theta$ . 2

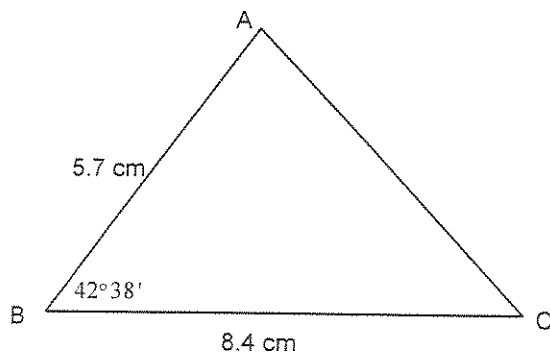
- c) Evaluate  $\lim_{x \rightarrow 2} \frac{x^2 - x - 2}{x - 2}$  1

- d) If  $\frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}} = x - y\sqrt{6}$  find the values of  $x$  and  $y$ . 3

- e) Calculate the shaded area 2



- f) Find the area of triangle  $ABC$  correct to 1 decimal place. 2



**Question 5** (Start a new page)

- a) Simplify  $\sqrt{98} - \sqrt{8} + \sqrt{12}$  1
- b) Simplify  $\frac{x+1}{x+2} \div \frac{x^2-9}{x^2+5x+6}$  2
- c) Find the equation of the line which passes through the point  $(1,4)$  3  
and is perpendicular to the line  $3x - 2y - 4 = 0$ , giving your  
answer in general form.
- d) From a point 26 metres from the base of a tower the angle of elevation 2  
of the top of the tower is  $68^\circ$ . Find the height of the tower,  
giving your answer correct to the nearest centimetre.
- e) Solve  $5 \sin \theta + 2 = 0$  for  $0^\circ \leq \theta \leq 360^\circ$  3  
( Give answers correct to the nearest degree )

**Question 6** (Start a new page)

- a) Find, correct to the nearest degree, the acute angle the line 2

$3x - y - 12 = 0$  makes with the  $x$  axis.

- b) Use the formula  $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$  2

to differentiate  $y = 3x^2 + x$  by first principles.

- c) A function is defined by

$$f(x) = \begin{cases} x^2 - 1 & \text{for } x \geq 1 \\ |x - 1| & \text{for } x < 1 \end{cases}$$

- i) evaluate  $f(4) + f(0)$  1

- ii) Sketch the function  $y = f(x)$  2

- d) A geologist drives 30 kilometres from point A on a bearing of

$324^\circ$  to point B. He then drives 50 kilometres from point B,

on a bearing of  $040^\circ$  to point C.

- i) Draw a diagram showing this information 1

- ii) Find the size of angle ABC. 1

- iii) Find the distance of point C from point A. 2

(Give answer correct to the nearest kilometre )

**Question 7** (Start a new page)

- a) How many sides has a regular polygon with interior angles equal to  $165^\circ$  ? 2
- b) Differentiate with respect to  $x$
- i)  $y = (2x + 1)(x - 3)$  1
- ii)  $y = \frac{1}{x} + x$  2
- c) Find the equation of the tangent to  $y = x^3 - 2x^2 - 3$  at the point  $(2, -3)$ . 3
- d) If  $f(x) = (x + 2)(x - 1)^4$  evaluate  $f'(2)$  3

**Question 8** (Start a new page)

a) Evaluate  $\lim_{x \rightarrow \infty} \frac{3x^2 - 2x - 1}{x^2 - 1}$  1

b) Prove the identity  $\tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta}$  3

c) Differentiate with respect to  $x$ .

i)  $y = \sqrt{x^2 - 5}$  1

ii)  $y = \frac{3x}{x-3}$  2

d) i) Find the gradient of the line  $2x + 5y + 6 = 0$  1

ii) The line  $ax + by + 11 = 0$  passes through the point  $(1, 4)$  3

and is parallel to the line  $2x + 5y + 6 = 0$ .

Find the values of  $a$  and  $b$ .

**End of Paper**



# SOLUTIONS

## QUESTION 1

a)  $(\frac{1}{2}, 3)$

b)  $a(12a^2 - 4a - 1)$   
 $= a(4a+1)(3a-1)$

c)  $12x - 3$

d)  $\sin 240^\circ = -\sin 60^\circ$   
 $= -\frac{\sqrt{3}}{2}$

e)  $x-3 = 2x+1$  or  $x-3 = -(2x+1)$   
 $x = -4$  or  $x-3 = -2x-1$   
 $3x = 2$   
 $x = \frac{2}{3}$

f)  $115\% \equiv \$207$   
 $\therefore 100\% \equiv \$180$

g)  $6x^2 - x = 0$   
 $x(6x-1) = 0$   
 $x = 0, \frac{1}{6}$

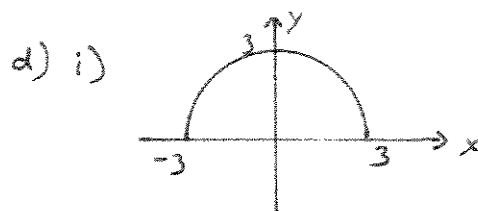
## QUESTION 2

a) 1.29

b)  $x - \frac{x+4}{4} = 2$   
 $4x - (x+4) = 8$   
 $3x - 4 = 8$   
 $3x = 12$   
 $x = 4$

c)  $\angle BFD = 52^\circ$  (alternate angles,  $AB \parallel CD$ )

$\therefore x + 95 + 52 = 180$  (straight angle)  
 $\therefore x = 33^\circ$



ii) Range:  $0 \leq y \leq 3$

e)  $2x - y = 7$   
 $x - 2y = -1$   
 $4x - 2y = 14$   
 $x - 2y = -1$   


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 $3x = 15$   
 $x = 5$   
 $\therefore y = 3$

~~subtract~~

## QUESTION 3

a) i)  $m_{AB} = \frac{5-1}{4-6}$   
 $= -2$

$\therefore y - 5 = -2(x - 4)$   
 $y - 5 = -2x + 8$   
 $2x + y - 13 = 0$

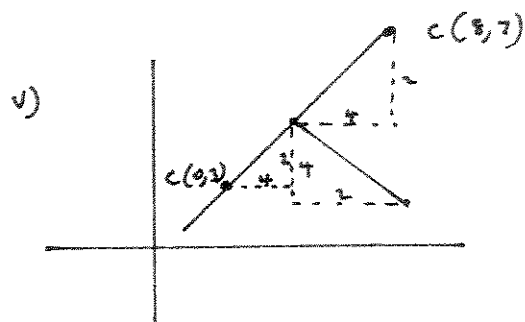
ii) when  $y = 0$   $x = 6\frac{1}{2}$   
 $\therefore T(6\frac{1}{2}, 0)$

$$\text{iii) } d = \sqrt{(4-6)^2 + (5-1)^2}$$

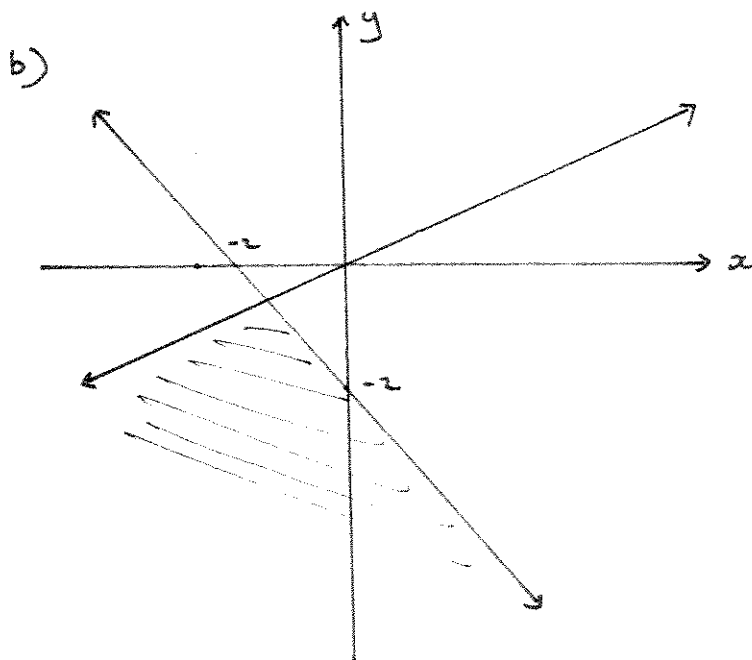
$$= \sqrt{20} \text{ units}$$

$$\text{iv) } d = \frac{|-1 \times 2 + 5 \times 1 - 13|}{\sqrt{2^2 + 1^2}}$$

$$= \frac{10}{\sqrt{5}} \text{ units}$$



$C(8, 7)$  or  $C(0, 3)$

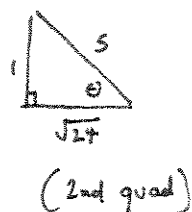


#### Question 4

a)  $y = 4$

b)  $\sin \theta = \frac{1}{5}$

$\cos \theta = -\frac{\sqrt{24}}{5}$



c)  $\lim_{x \rightarrow 2} \frac{(x-2)(x+1)}{x-2}$

$$= 3$$

d)  $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}} \times \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}}$

$$= \frac{(\sqrt{3}-\sqrt{2})^2}{1}$$

$$= 3 - 2\sqrt{6} + 2$$

$$= 5 - 2\sqrt{6}$$

$\therefore x = 5, y = 2$

e)  $A = 10 \times 4$

$$= 40 \text{ sq units}$$

f)  $A = \frac{1}{2} \times 8.4 \times 5.7 \times \sin 42^\circ 38'$

$$= 16.2 \text{ cm}^2$$

#### Question 5

a)  $7\sqrt{2} - 2\sqrt{2} + 2\sqrt{3}$

$$= 5\sqrt{2} + 2\sqrt{3}$$

b)  $\frac{x+1}{x+2} \times \frac{(x+3)(x+2)}{(x-3)(x+3)}$

$$= \frac{x+1}{x-3}$$

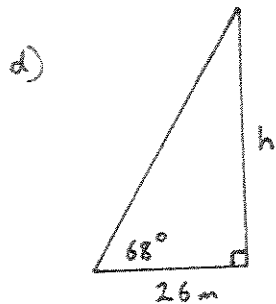
c)  $3x - 2y - 4 = 0$

$$m = \frac{3}{2}$$

$\therefore m_{\perp} = -\frac{2}{3}$

$\therefore y - 4 = -\frac{2}{3}(x - 1)$

$$2x + 3y - 14 = 0$$



$$\begin{aligned}\tan 68^\circ &= \frac{h}{26} \\ h &= 26 \times \tan 68^\circ \\ &= 64.35\text{m}\end{aligned}$$

e)  $5 \sin \theta + 2 = 0$

$$\sin \theta = -\frac{2}{5}$$

(3rd / 4th  $\Rightarrow \theta = 24^\circ$ )

$$\therefore \theta = 204^\circ, 336^\circ$$

#### QUESTION 6

a)  $m = 3$

$$\therefore \tan \theta = 3$$

$$\therefore \theta = 72^\circ$$

b)  $y' = \lim_{h \rightarrow 0} \frac{3(x+h)^2 + (x+h) - [3x^2 + x]}{h}$

$$= \lim_{h \rightarrow 0} \frac{3x^2 + 6xh + 3h^2 + x + h - 3x^2 - x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{6xh + 3h^2 + h}{h}$$

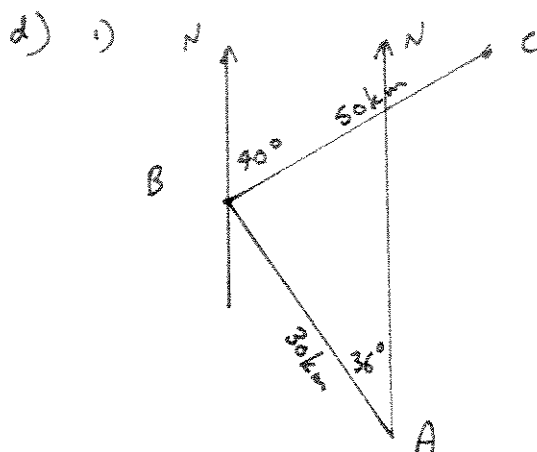
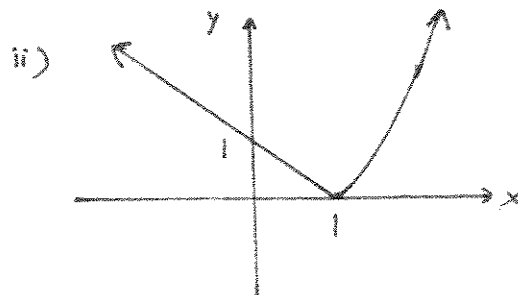
$$= \lim_{h \rightarrow 0} 6x + 3h + 1$$

$$= 6x + 1$$

c) i)  $f(4) + f(0)$

$$= 15 + 1$$

$$= 16$$



ii)  $\angle ABC = 180^\circ - 36^\circ - 40^\circ$   
 $= 104^\circ$

iii)  $AC^2 = 30^2 + 50^2 - 2 \times 30 \times 50 \times \cos 104^\circ$

$$\therefore AC = 64 \text{ km}$$

#### QUESTION 7

a) ext angle  $= 15^\circ$

$$\therefore \text{no. of sides} = \frac{360}{15}$$

$$= 24$$

b) i)  $y = 2x^2 - 5x - 3$

$$\therefore y' = 4x - 5$$

ii)  $y = x^{-1} + x$

$$\therefore y' = -x^{-2} + 1$$

$$c) \quad y' = 3x^2 - 4x$$

when  $x=2$

$$m_T = 3 \times 2^2 - 4 \times 2$$

$$= 4$$

$$\therefore y+3 = 4(x-2)$$

$$y+3 = 4x-8$$

$$y = 4x-11$$

$$d) \quad f(x) = (x+2)(x-1)^4$$

$$f'(x) = (1)(x-1)^4 + (x+2)4(x-1)^3$$

$$\therefore f'(2) = 1^4 + 4 \times 4 \times 1^3$$

$$= 17$$

### QUESTION 8

$$a) \quad 3$$

$$b) \quad LHS = \tan \theta + \cot \theta$$

$$= \frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$$

$$= \frac{\sin^2 \theta + \cos^2 \theta}{\cos \theta \sin \theta}$$

$$= \frac{1}{\cos \theta \sin \theta}$$

$$= RHS$$

$$c) \quad i) \quad y = (x^2-5)^{\frac{1}{2}}$$

$$\therefore \frac{dy}{dx} = \frac{1}{2} (x^2-5)^{-\frac{1}{2}} \cdot 2x$$

$$= x (x^2-5)^{-\frac{1}{2}}$$

$$= \frac{x}{\sqrt{x^2-5}}$$

$$ii) \quad \frac{dy}{dx} = \frac{(x-3)(3) - (3x)(1)}{(x-3)^2}$$

$$= \frac{-9}{(x-3)^2}$$

$$d) \quad i) \quad m = -\frac{2}{5}$$

$$ii) \quad \text{gradient of } ax+by+11=0$$

$$m = -\frac{a}{b}$$

$$\text{as parallel} \quad -\frac{a}{b} = -\frac{2}{5}$$

$$\therefore 5a = 2b \quad *$$

$$\text{sub } (1,4) \text{ into } ax+by+11=0$$

$$a + 4b + 11 = 0 \quad *$$

Solve simultaneously (\*)

$$5a = 2b$$

$$a + 4b + 11 = 0$$

$$\therefore a + 2 \times 5a + 11 = 0$$

$$11a + 11 = 0$$

$$a = -1$$

$$\therefore b = -\frac{5}{2}$$

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